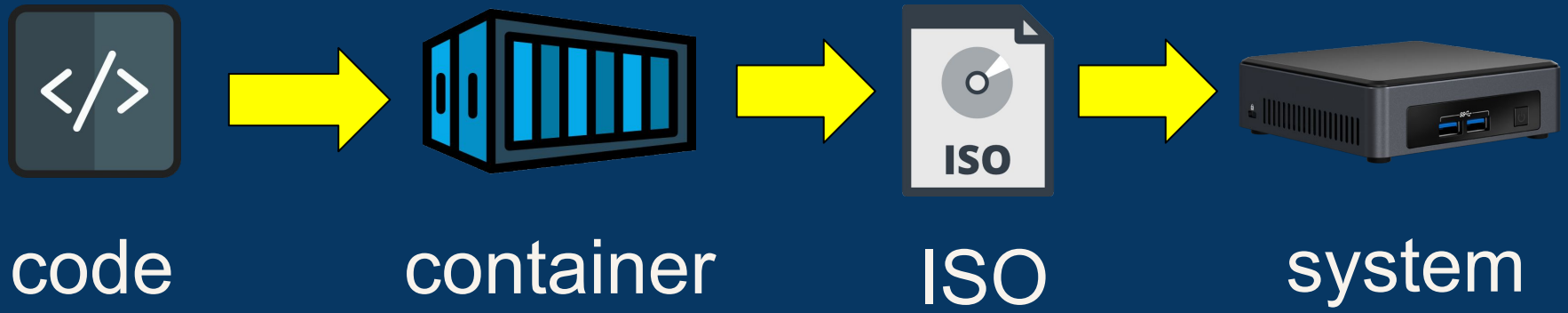
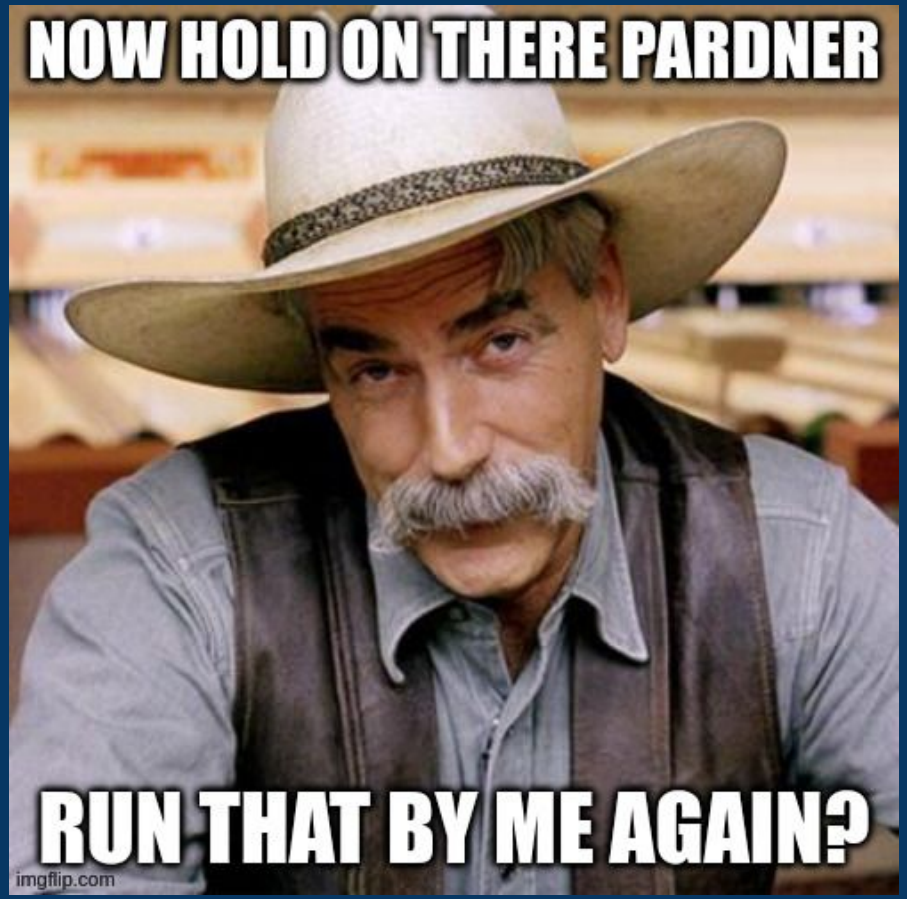
How to provision your hardware with … a container?

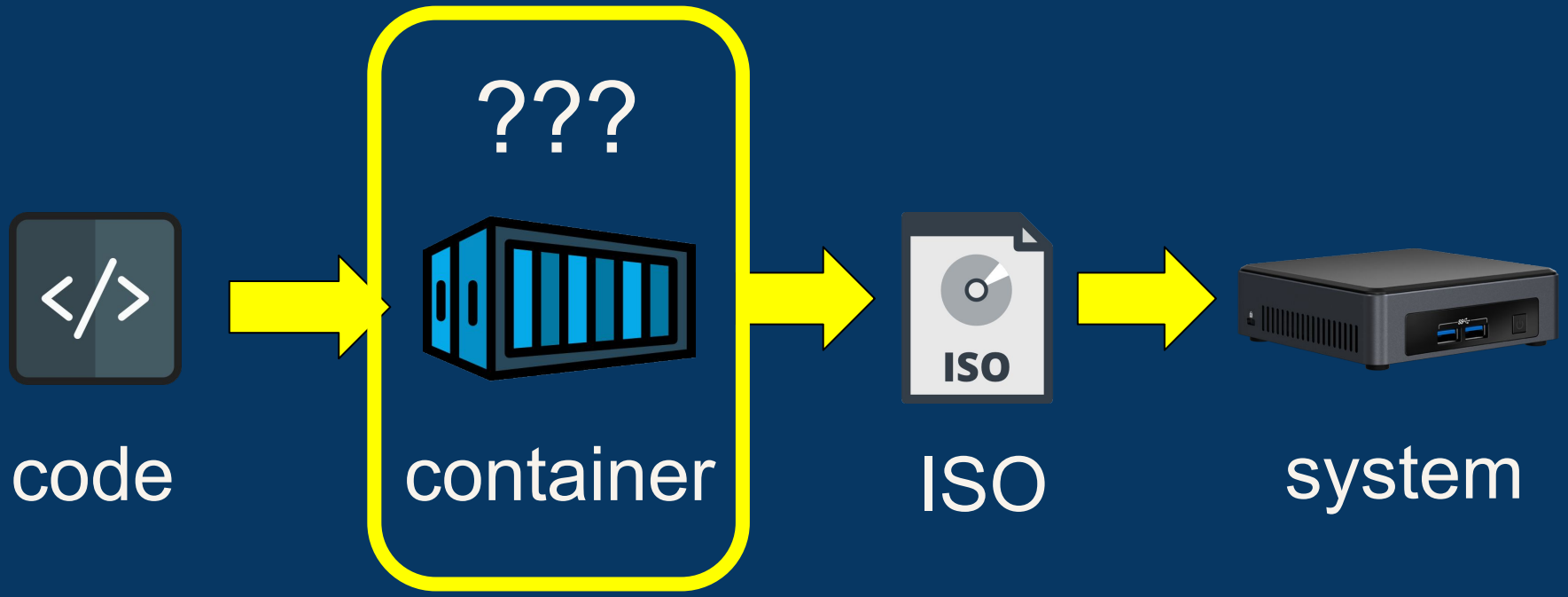
Josh Berkus

Red Hat

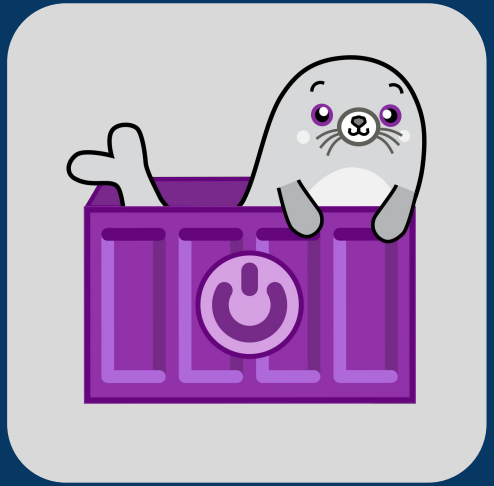
SCALE 23x


ISO
code
container
ISO
system


NOW HOLD ON THERE PARDNER
RUN THAT BY ME AGAIN?
imgflip.com


???
ISO
code
container
ISO
system

A brief digression into the history of OS deployment tech

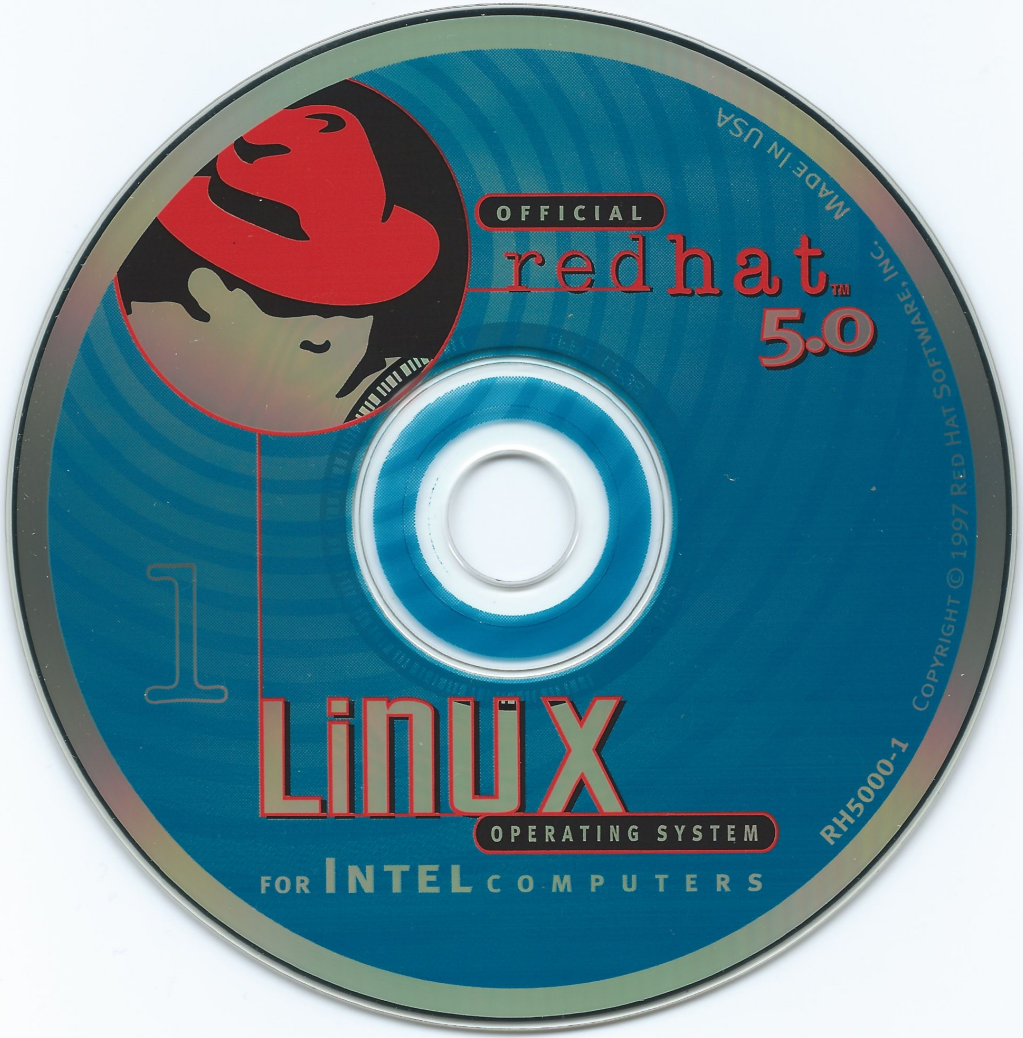


1995: Compile Everything

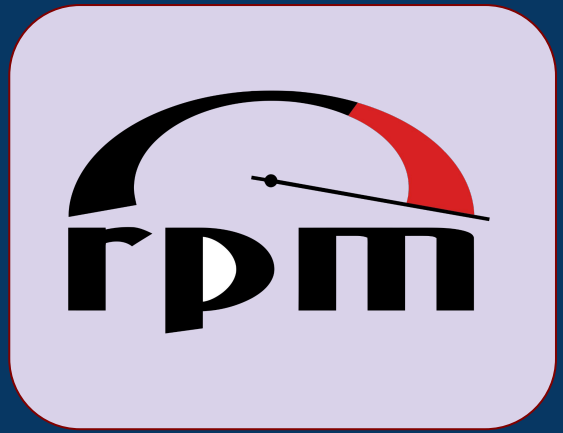
```
Linux Kernel Configuration
Arrow keys navigate the menu.  <Enter> selects submenus --->.
Highlighted letters are hotkeys.  Pressing <Y> includes, <N> excludes,
<M> modularizes features.  Press <Esc><Esc> to exit, <?> for Help.
Legend: [*] built-in  [ ] excluded  <M> module  < > module capable

    Code maturity level options  --->
    General setup  --->
    Loadable module support  --->
    Processor type and features  --->
    Power management options (ACPI, APM)  --->
    Bus options (PCI, PCMCIA, EISA, MCA, ISA)  --->
    Executable file formats  --->
    Device Drivers  --->
    File systems  --->
```

1998: Packages

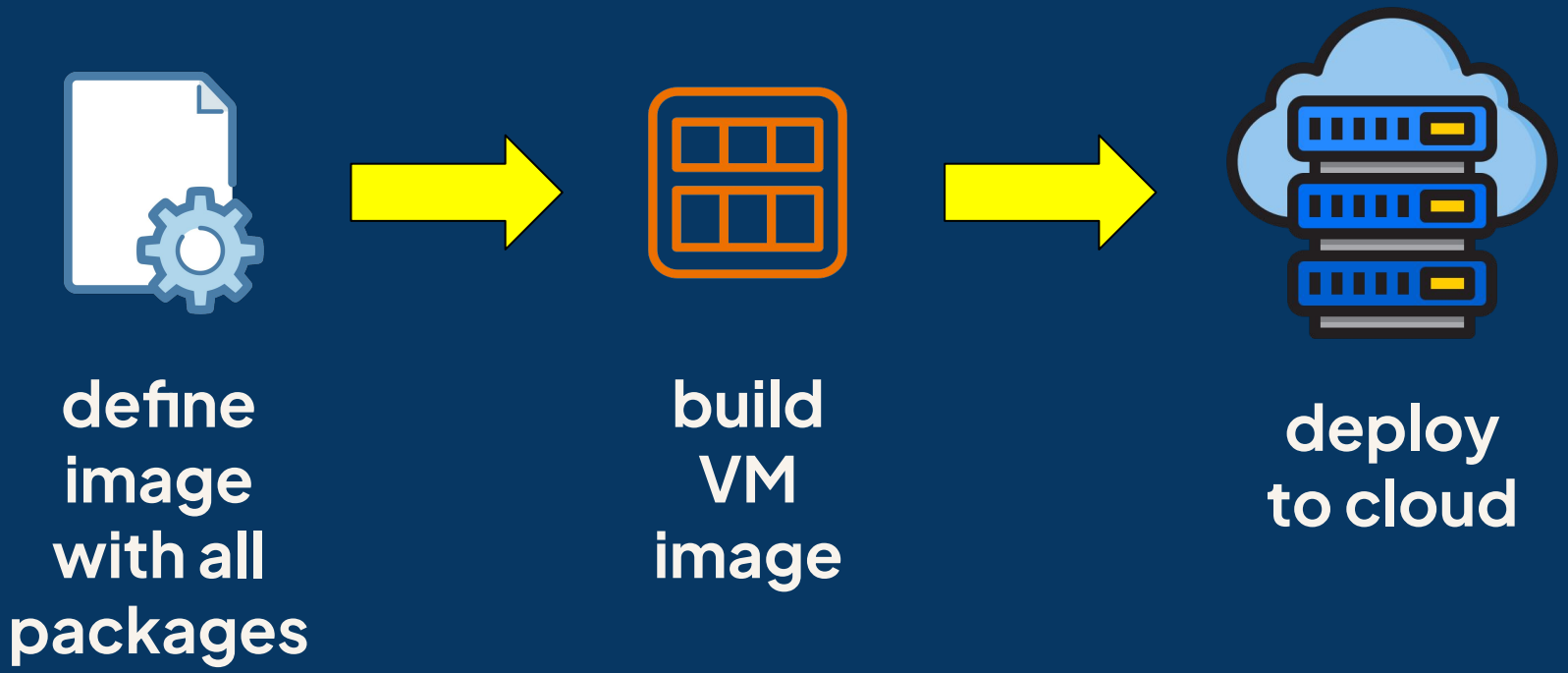


2005: Packages & ConfigMgmt



OS Install Scaling Problems

- Too slow for large numbers of machines
- Non-determinative outcomes
- Failures can't self-heal
- Hard to parallelize
- No rollbacks



2010: Deploying "Images"

VM Image Problems

Difficult composition tools

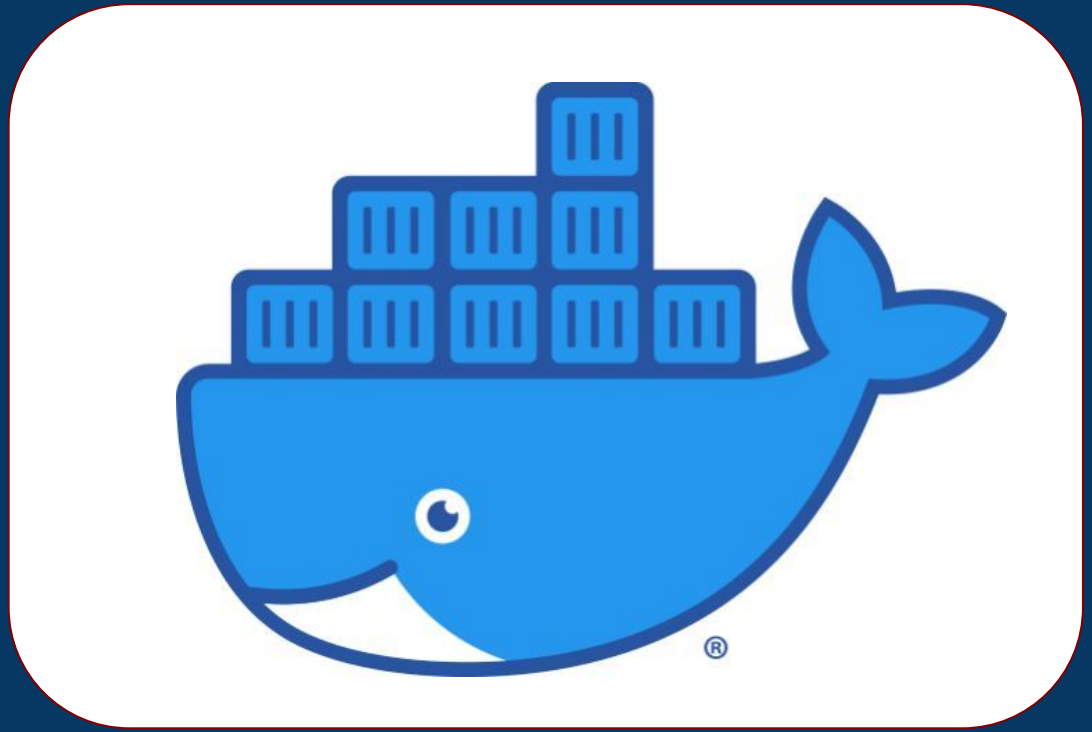
Images are slow to build

Large images are hard to distribute

Works poorly with bare metal

Still No Rollbacks

2014: Application Containers



Container Image Advantages

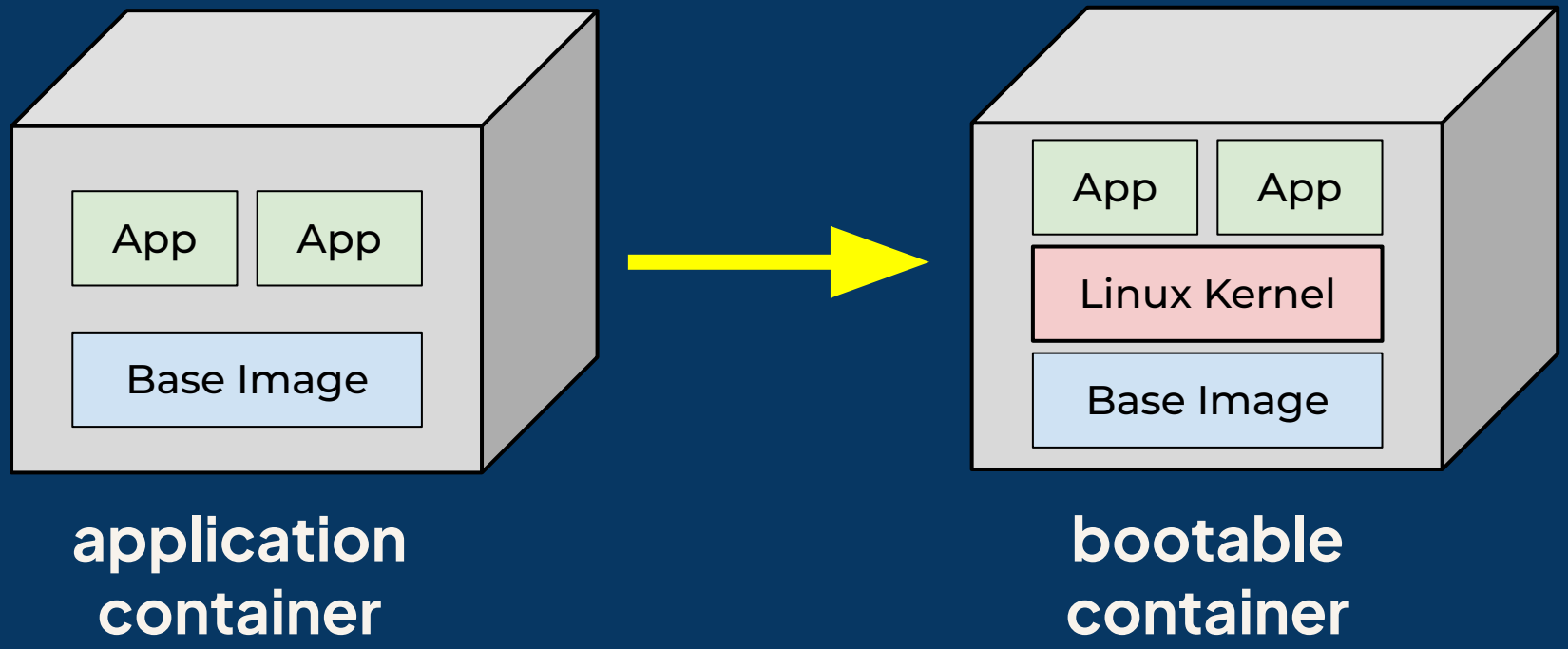
- Lots of good, familiar build tools
- Images build faster
- Smaller images in container registries
- Deploy on bare metal *and* cloud
- Rollbacks!

Introducing bootc

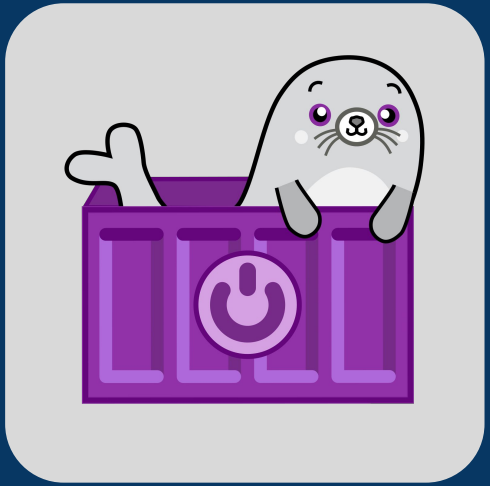
bootc: bootable containers

- builds bootable container images
- deploys containers as system images
- manages system status and updates
- supports rollback

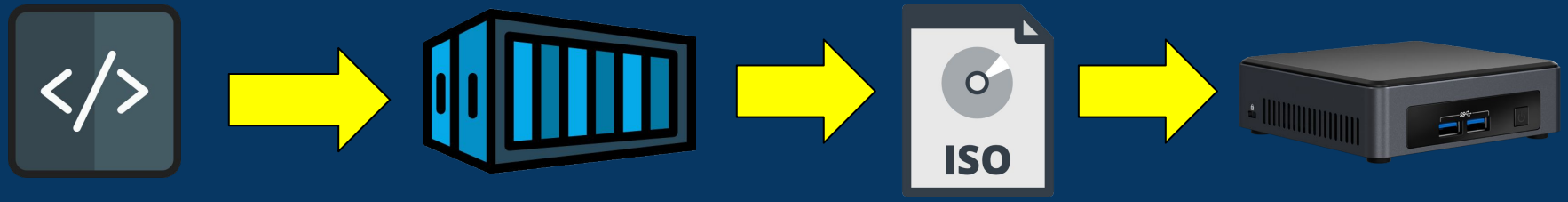
bootc: bootable containers



And now, back to building our system

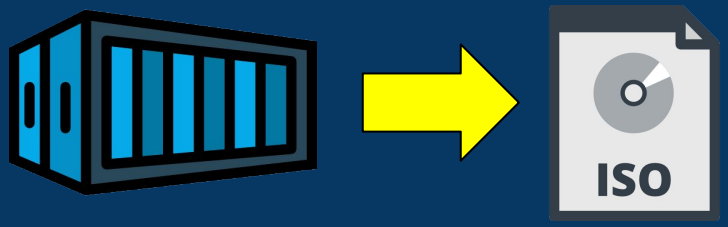


initial provisioning



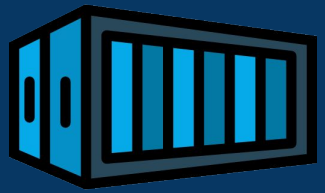
code container ISO system

making isos



container ISO

making isos



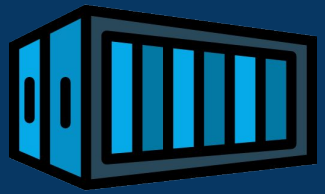
container

bootc image builder

ISO

Y U no demo?

code

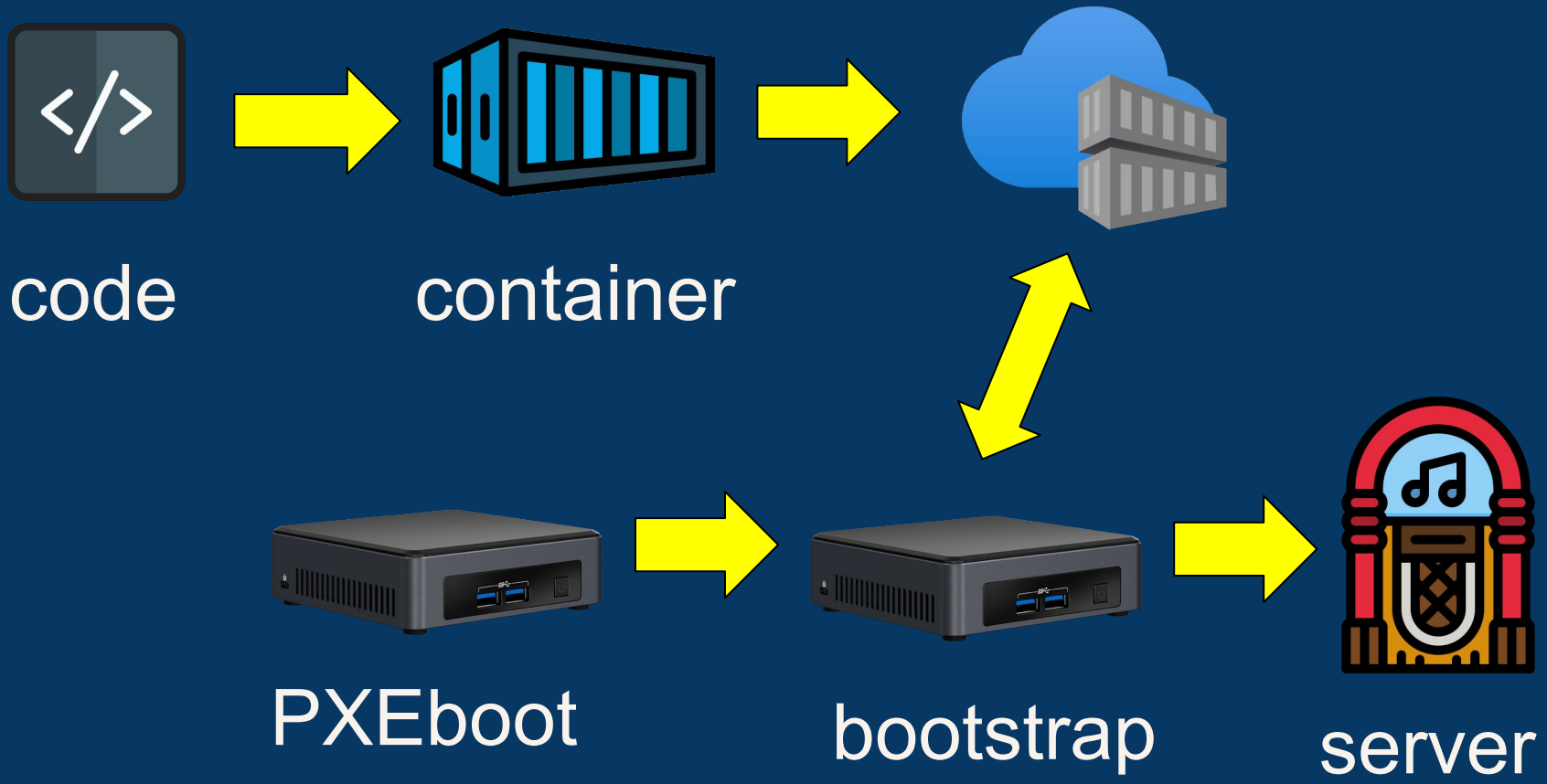


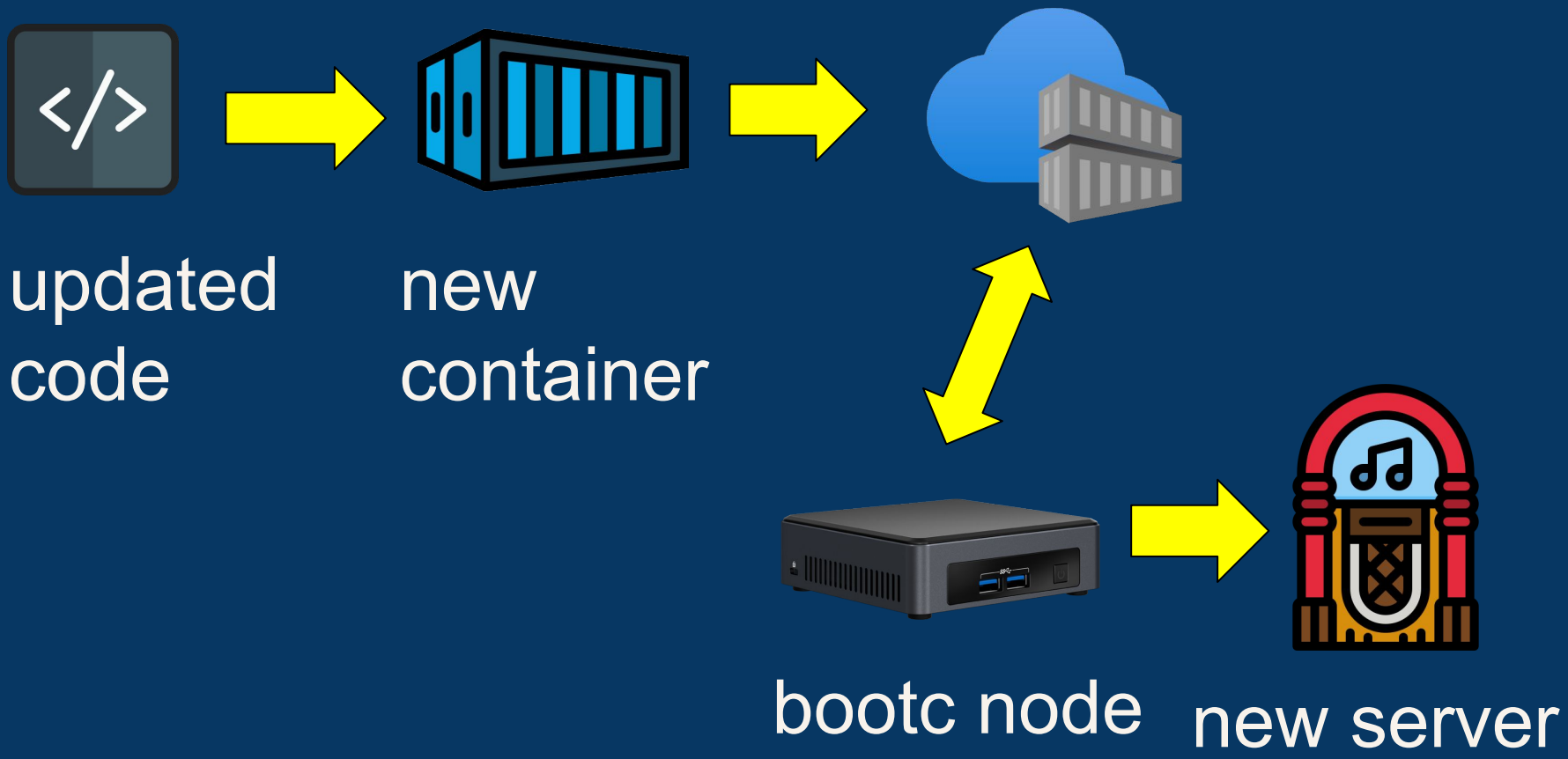
container

ISO

system

Checking storage configuration...
Installation
1) Ex1 Language settings
(Regional Setting Status)
2) Ex1 Keyboard Settings
(Keyboard Layout)
3) Ex1 Time settings
(Europe/Los_Angeles timezone)
4) Ex1 Network configuration
(Connected: eapinf)
5) Ex1 Disk partitioning
(Automatic partitioning selected)
6) Ex1 User creation
(No user will be created)
Progress
Setting up the installation environment
Configuring storage
Creating disklabel on /dev/sda
Creating sda1 on /dev/sda1
Creating sda2 on /dev/sda2
Creating sda3 on /dev/sda3
Creating sda4 on /dev/sda4
Running pre-installation scripts
Running pre-installation tasks
Installing the software
Begin/apt/ stating: /var/install/repo/installapp
WARNING
Beelink
intel
312-10112-01


code
container
PXEboot
bootstrap
server


updated code
new container
bootc node
new server

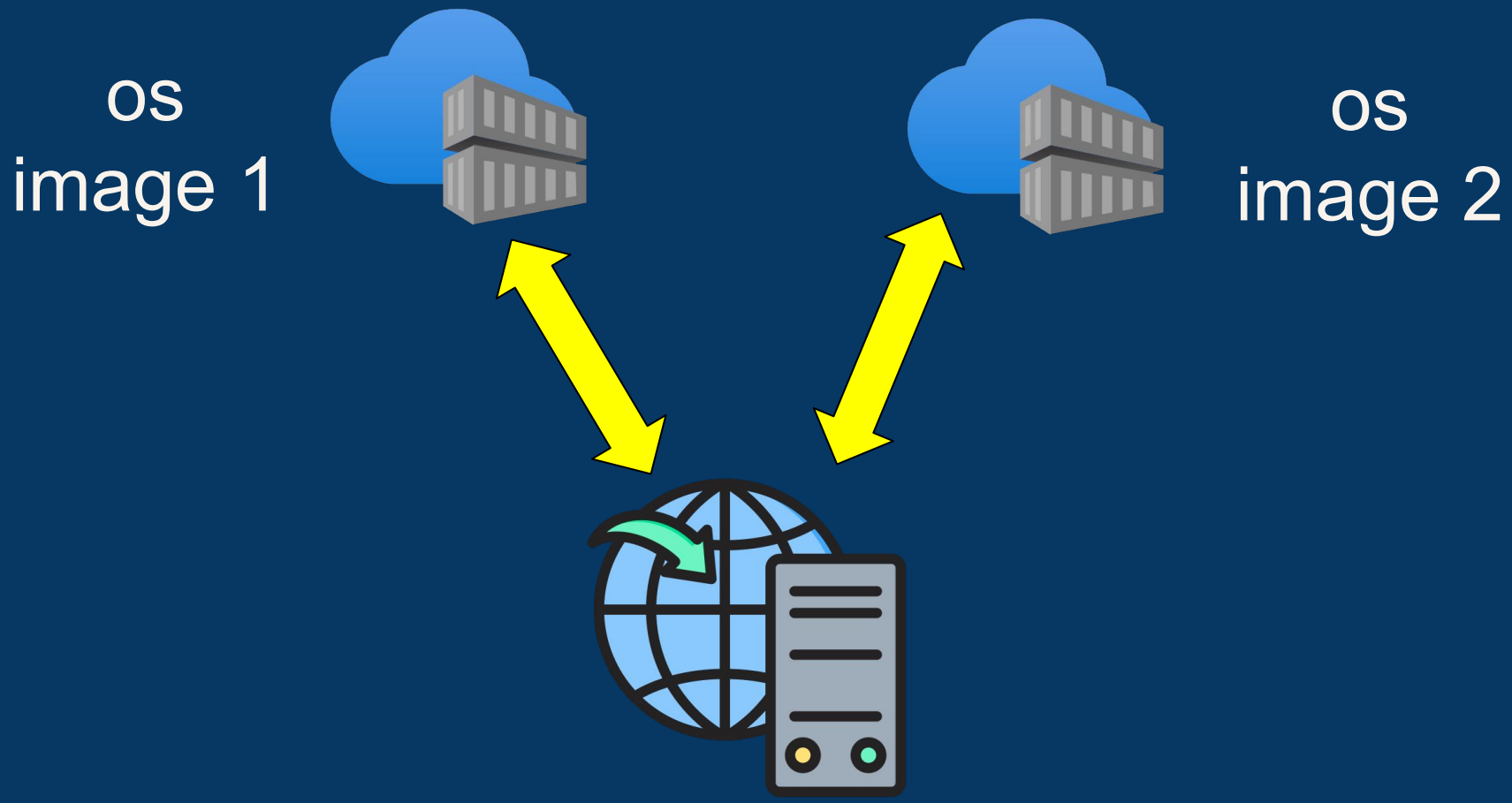
Demo code:

github.com/jberkus/bootc-device-demo

bootc update advantages

- No need to choose components
- Updates by efficient image layering
- No possibility of a partial update
- Rollback if update fails
- bootc switch

bootc switch?


os
image 1
os
image 2

os
image 1

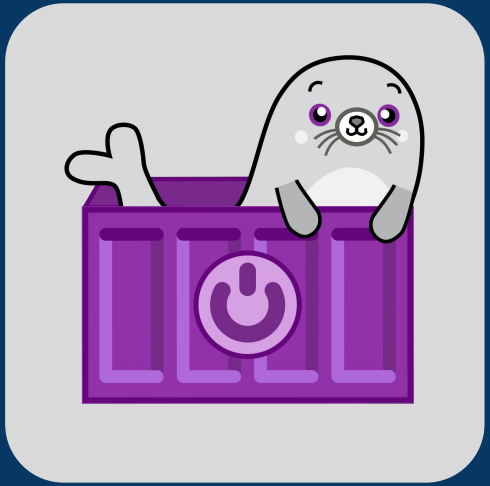
os
image 2

Not Just For Edge Devices

fedora ATOMIC DESKTOPS

bazzite

What's next?



My jukebox distro

Generic bootc OS image

Quadlet autostart

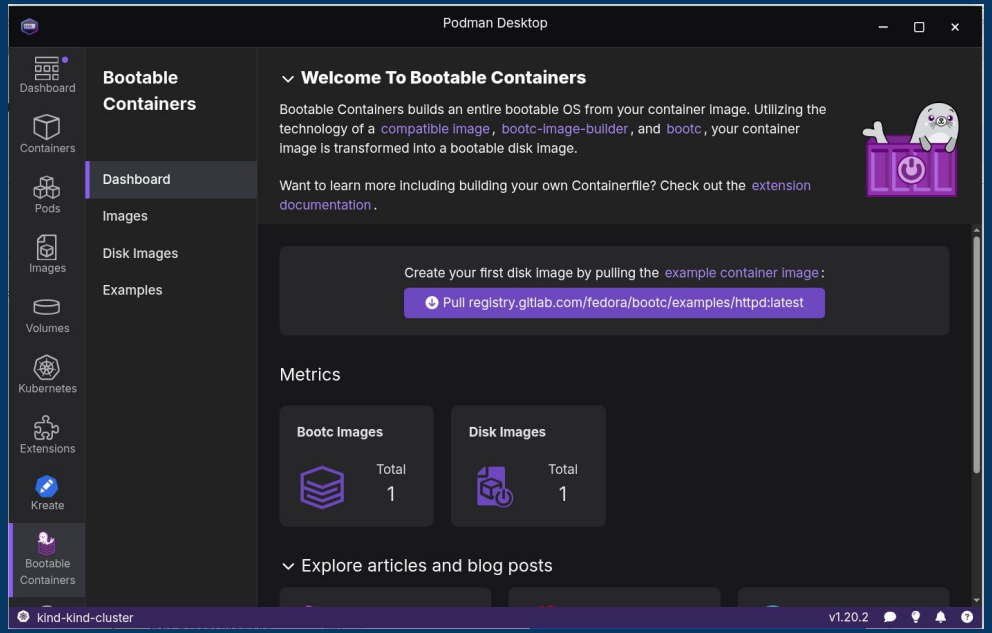
Preloaded containers

Media player: Plex? OSMC? Kodi?

Music storage?

Getting Started with bootc

1. Download **Podman Desktop** for your platform (Win/Mac/Lin)
2. Install the bootc extension
3. Play around with creating your own bootable images



Join the bootc community!

- bootc-dev.github.io
- Slack & Matrix chat, Friday dev meetings
- You can contribute!
  - Bug reports
  - Platform tests
  - Documentation

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